Measuring the Charge-diffusion Kernel of a Back-illuminated CCD in Orbit from Glancing Particle Tracks

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Abstract.
The lateral diffusion of photogenerated charge in the undepleted layer of a back-illuminated silicon sensor sets its point spread function and the statistics of its noise in the ultraviolet, but it is rarely measured on a flight sensor. We show that glancing particle tracks, which cross the full thickness of the sensor over a length of tens of pixels, sample the diffusion width at every depth in a single exposure, and we use them to measure the depth-dependent charge-diffusion kernel of the four CCDs of the Interface Region Imaging Spectrograph in orbit, from 10044 tracks found in level-1 images taken mostly inside the South Atlantic Anomaly. The width of the charge cloud at the back surface, the depth at which the field-free layer ends, the spread acquired drifting across the depletion region, and the probability that two electrons deposited at the same depth are collected in the same pixel are measured directly on each sensor. The tracks are fit better by the charge cloud of diffusion between a reflecting back surface and the depletion region than by a Gaussian, and the width of that cloud at the back surface implies sensors 11.0 to 12.8 μm thick rather than the 14 μm of the design. The field-free layer ends at 0.40 of the thickness on the Slit-Jaw Imager, where the cloud at the back surface is 5.0 μm wide and the same-pixel probability is 0.348 ± 0.004, and the tracks require a further spread of 0.80 μm inside the depletion region. The four sensors, of one design, differ by a fifth in the width of the cloud at the back surface. The method needs no laboratory access and applies to any back-illuminated sensor in orbit.

Keywords: charge-coupled devices, charge diffusion, point spread function, ultraviolet detectors, particle tracks, IRIS.
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1 Introduction

A back-illuminated silicon charge-coupled device (CCD) absorbs ultraviolet (UV) light within a fraction of a micron of its back surface, in a layer where there is no electric field to sweep the photogenerated charge toward the gates. The charge diffuses until it reaches the edge of the depletion region, and the resulting charge cloud, which can be several microns wide, is shared among neighboring pixels. This sharing broadens the point spread function (PSF) of the sensor[1] and, because the electrons liberated by one photon are shared together, it changes the statistics of the image noise, entering the variance-to-mean ratio (VMR) of a photon-transfer curve through the probability that two electrons from the same photon land in the same pixel.[2]

The width of the charge cloud is set by the thickness of the undepleted layer, which depends on the resistivity of the wafer and the bias applied to the device, both of which vary between nominally identical sensors and are rarely known for a sensor in orbit. Laboratory measurements use light of varying penetration depth[3,4] or the split events of soft X-rays,[5] and neither is available once the instrument has launched.

Energetic particles offer a probe that arrives for free. A particle crossing the sensor at glancing incidence deposits charge along a track tens of pixels long, and because it enters at the back surface and exits at the gates, the fractional position along the track is the fractional depth of the charge. Cosmic-ray muons have been used in this way to measure the growth of charge diffusion with drift distance in thick, fully depleted CCDs,[6] where the cloud widens with depth. In a thin sensor with an undepleted back surface the dependence is reversed, the cloud is widest at the back surface and

collapses to a pixel-sharp line in the depletion region, and the same tracks measure the field-free kernel that governs UV imaging.

Here we apply this method to the four CCDs of the Interface Region Imaging Spectrograph (IRIS),[7] whose far ultraviolet (FUV) and near-UV photon-transfer curves[8] motivated the question. Section 2 describes the tracks, the model of their cross-section, and the fit; Section 3 presents the kernel of each sensor; and Section 4 discusses the systematics, including the charge density of the tracks.

2 Method

2.1 Data

IRIS[7] carries four CCDs of the same design: two behind the FUV channel of the spectrograph (FUV1 and FUV2, which share one level-1 image), one behind its near ultraviolet (NUV) channel, and one behind the Slit-Jaw Imager (SJI). They are back-illuminated, 14 μm thick, with 13 μm pixels, and their level-1 images are the raw frames with only the dark and the readout corrections applied, so a particle hit appears exactly as the camera recorded it. We searched 23792 such images, 8260 of them taken inside the South Atlantic Anomaly (SAA), where the flux of trapped protons at the altitude of IRIS is greatest,[9,10] drawn from the campaigns of Table 1. The campaigns added for this work were chosen among those exposing for 8 s, since a longer exposure crowds the frame inside SAA with hits that spoil the tracks around them (the campaigns of the original measurement that pass through SAA expose for 15 s, and yielded per frame half as many of the tracks kept below), and among those whose spectrograph readout window lies mostly beyond the limb, as the pointing recorded in each frame's header places it. Figure 1 shows a spectrograph image and a slit-jaw image taken seconds apart during one pass through SAA: the particle hits are dense, and a handful of them are the long glancing tracks we want. Tracks were sought only where the pointing recorded in each frame's header places the pixel at least 15 arcseconds above the photospheric limb, so that no track lies on the disk or in the chromosphere and spicules seen just above the limb.

2.2 Finding tracks

The background of each frame was estimated as the trimmed mean of the frames in the same observing sequence and subtracted, and the read noise was estimated from the median absolute deviation of the residual. Pixels more than five times the read noise above the background were labeled, and a straight line was fit by least squares to the charge-weighted centroid of each row of every connected group of labeled pixels. We kept the groups spanning at least twelve rows (or columns) with a slope below 0.35 pixels per row, so that each track is nearly aligned with the pixel grid and its cross-section is sampled once per row without resampling the pixels, and extracted a seven-pixel-wide cutout centered on the integer part of the fitted line in every row. This yielded 10044 tracks. The slope cut is not as costly as it sounds: Figure 2 shows that the trapped protons arrive strongly aligned with one axis of the detector or the other, which one depending on the roll of the spacecraft, so that more of them pass it than the 43% an isotropic population would, 57% at the roll of most of the campaigns.

Table 1 The observing campaigns searched for tracks, with the cameras kept from each (F for the far-ultraviolet spectrograph, N for the near-ultraviolet one, S for the slit-jaw imager), the spacecraft roll, the exposure time, and the number of frames searched, with the number taken inside SAA in parentheses. The tracks of every campaign but the two marked †, whose frames were taken outside SAA, are left by trapped protons; those two hold cosmic rays. The core tracks are the flat tracks with $0.25 < t_c < 0.6$, over which the mean t_c with its standard error and the median $\sigma_{\max}$ with its interquartile range are taken.

date	cameras	roll	exp. (s)	frames	tracks	flat	core	t_c	$\sigma_{\max}$ (μm)
2014-03-12	FN	0°	15	888 (114)	608	189	158	0.391 ± 0.006	4.5 [4.0, 5.0]
2014-04-07†	FN	0°	4	1950 (0)	34	16	14	0.396 ± 0.016	5.2 [4.6, 5.5]
2018-03-16†	FN	0°	15	622 (34)	118	56	46	0.365 ± 0.009	5.0 [4.1, 5.0]
2018-05-04	FN	−90°	15	722 (273)	611	183	148	0.396 ± 0.005	4.5 [4.0, 5.5]
2018-05-04/06	S	−90°	15	1156 (469)	471	165	143	0.401 ± 0.005	5.0 [4.5, 5.5]
2014-08-20	FNS	−75°	8	782 (333)	636	334	299	0.388 ± 0.003	5.0 [4.5, 5.5]
2014-08-21	FNS	−75°	8	768 (318)	800	383	348	0.387 ± 0.003	4.5 [4.0, 5.5]
2014-08-23	FNS	90°	8	1214 (333)	800	484	443	0.392 ± 0.003	5.0 [4.5, 5.5]
2014-08-24	FNS	90°	8	806 (330)	942	494	431	0.395 ± 0.003	5.0 [4.5, 5.5]
2014-08-25	FNS	90°	8	778 (321)	1482	651	589	0.393 ± 0.002	5.0 [4.5, 5.5]
2014-11-02	FNS	0°	8	762 (300)	1153	482	411	0.387 ± 0.003	5.0 [4.0, 5.0]
2014-11-27	FNS	0°	8	970 (383)	144	9	7	0.357 ± 0.019	4.5 [4.5, 5.0]
2015-03-26	FNS	−90°	8	616 (210)	65	34	31	0.402 ± 0.012	5.0 [4.5, 5.0]
2015-09-20	FNS	0°	8	1039 (575)	505	261	233	0.388 ± 0.004	5.0 [4.5, 5.5]
2016-01-02	FNS	0°	8	3836 (1134)	542	242	206	0.394 ± 0.004	4.5 [4.0, 5.0]
2016-01-06	FN	0°	8	638 (184)	90	68	66	0.387 ± 0.008	4.5 [3.5, 5.0]
2016-05-16	FNS	0°	8	1174 (732)	511	240	209	0.386 ± 0.004	5.0 [4.5, 5.5]
2016-10-28	FN	0°	8	556 (234)	186	95	81	0.399 ± 0.007	4.5 [4.5, 5.0]
2016-12-13	FNS	0°	8	1737 (990)	123	79	74	0.382 ± 0.007	4.5 [4.0, 5.0]
2016-12-17	FNS	0°	8	1247 (192)	3	2	1	0.350 ± 0.000	5.5 [5.5, 5.5]
2016-12-25	FNS	0°	8	1531 (801)	220	117	107	0.379 ± 0.006	4.5 [4.2, 5.0]

2.3 Model

Figure 3 shows the principle. A particle that enters the back surface and exits the front surface spans the full thickness of the sensor, so the fractional depth of slice i of an N-slice track is $t = (i + 1/2)/N$, up to the orientation of the track, and the charge deposited in each slice diffuses from a known depth. We model the charge in every slice as spread uniformly along the fitted centerline and diffused by a charge cloud with two parts.

The first part is the field-free layer, which the charge of a slice crosses only if $t < t_c$, where $t_c = z_f/D$ is the fractional thickness of the layer. Charge created there diffuses until it reaches the edge of the depletion region, which collects it, while the back surface reflects it.[1] In a layer z_f thick, the fraction of the charge created a fraction t/t_c of the way across it that arrives at an offset less than y along one axis is

$$F(y) = \frac{1}{2} + \frac{1}{\pi} \arctan \left[\frac{\sinh(\pi y/2z_f)}{\cos(\pi t/2t_c)} \right], \tag{1}$$

whose variance falls from z_f^2 at the back surface to zero at the depletion region as $z_f^2[1 - (t/t_c)^2]$. The cloud is not a Gaussian: it falls off exponentially, as $e^{-\pi|y|/2z_f}$, and near the depletion region, where most of the charge is collected at once but some wanders back toward the back surface first, it has a narrow core and broad wings. Equation 1 ties the width of the cloud at the back surface to the thickness of the layer, $\sigma_{\max} = z_f = t_c D$. We leave $\sigma_{\max}$ free instead, replacing z_f by it

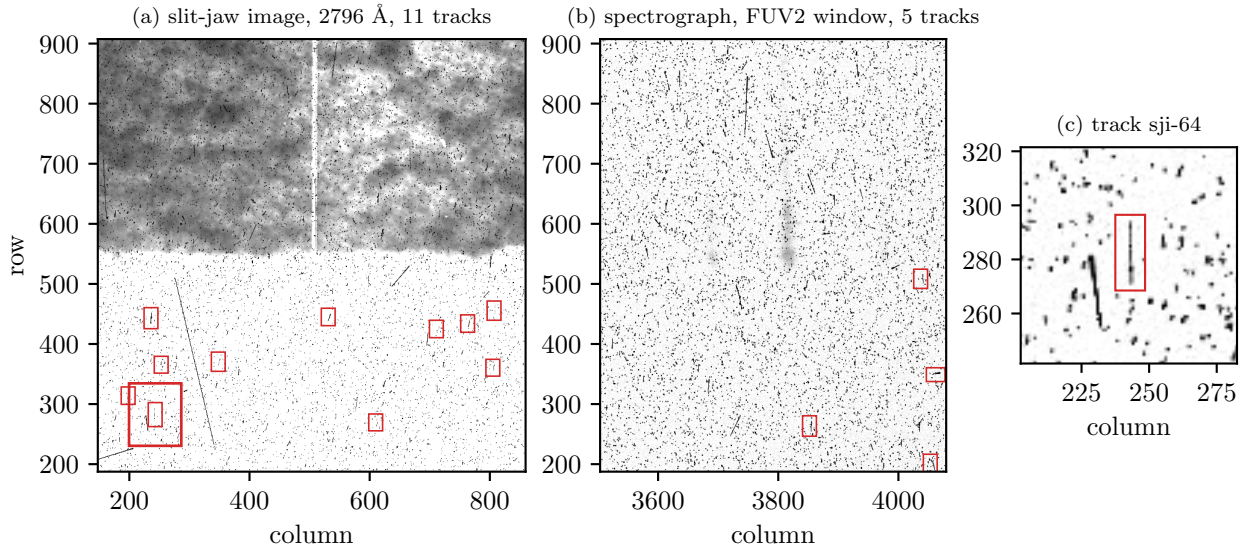


Fig 1 Level-1 images taken seconds apart during a pass through SAA on 2018 May 4, with the spacecraft rolled by $-90°$ so that the limb runs horizontally across the field. (a) The slit-jaw image at 2796 Å, whose lower half is off the limb. (b) The widest of the readout windows of the spectrograph image, on the FUV2 CCD. (c) One of the tracks in (a) at full resolution. The images are shown with an inverse hyperbolic sine stretch above the background, saturating at 400 data numbers, and the red boxes outline the tracks the finder kept from each frame. Most of the particle hits are the short streaks and dots that pepper both images; the tracks used here are the long, thin, nearly vertical ones.

in Equation 1, so that the model has the same parameters as the Gaussian model below, and the ratio $\sigma_{\max}/t_c$ measures the thickness of the sensor the fit implies. The second part is the depletion region. Charge that reaches the edge of the depletion region, whether it was deposited there or diffused there from the field-free layer, still has to drift to the gates and spreads a little as it does so, which we model as a Gaussian of variance $\sigma_d^2\, g(t)$. σ_d is the spread acquired crossing the full depleted thickness, and $g(t) = \min[(1-t)/(1-t_c), 1]$ scales it with the distance drifted, so that it is one for all charge deposited in the field-free layer and falls linearly to zero at the gates. The variance of the cloud is then

$$\sigma^2(t) = \sigma_{\max}^2 \left[1 - \left(\frac{t}{t_c}\right)^2\right] + \sigma_d^2\, g(t), \tag{2}$$

where the first term applies only for $t < t_c$. Equation 1 convolved with the Gaussian has no closed form, so we evaluate the cloud as the mixture of eight Gaussians of the `optika` package,[11] whose variances scale as $\sigma_{\max}^2$ and to each of which we add $\sigma_d^2\, g(t)$; at a single depth it reproduces Equation 1 to about a part in a thousand of the charge. We work in fractional depth because we observe the length of a track in slices rather than the thickness of the sensor in microns, and the uniform spread along the tilted centerline within a slice is approximated by adding $m^2/12$ to the variance of every Gaussian, where m is the slope in pixels per row.

To see what the shape of the cloud is worth, we also fit every track with a Gaussian whose variance is

$$\sigma^2(t) = \sigma_{\max}^2 \left(1 - \frac{t}{t_c}\right) + \sigma_d^2\, g(t), \tag{3}$$

the depth dependence of the model of Janesick,[2] a random walk to an absorbing plane at z_f with no reflecting back surface, with the same parameters searched on the same grids.

The three parameters are not fit alike. t_c and $\sigma_{\max}$ describe the field-free layer, whose thickness can vary across a wafer, and are fit to each track together with its orientation, by exhaustive search on a grid of 21 values in each of $t_c \in [0, 1]$ and $\sigma_{\max} \in [0, 10]$ μm. At each grid point the offset and tilt of the centerline are adjusted, by up to 0.6 pixels and 0.03 pixels per row, to minimize a robust misfit,

$$M = \sum_i \sum_j \ln\left(1 + \frac{r_{ij}^2}{2}\right), \qquad r_{ij} = \frac{f_{ij} - q_{ij}}{\epsilon_i}, \tag{4}$$

where f_{ij} is the fraction of the charge of slice i measured in pixel j of the cutout, q_{ij} the fraction the model puts there, and ϵ_i the read noise of a pixel divided by the charge of the slice, the uncertainty of a fraction from read noise alone, so that the brighter slices count for more. This is the negative log-likelihood of a Cauchy distribution rather than a Gaussian, so that a single stray pixel, from a delta ray or a second hit inside the cutout, cannot dominate the fit, and differences in M are what the thresholds below are stated in. σ_d describes the drift field, which is a property of the CCD and its bias rather than of any one track, and a single track constrains it only weakly, so it is fit to each CCD as a whole. Every track is fit as above at each of 16 values of σ_d from 0 to 1.5 μm, the misfits M of the flat tracks, defined next, are summed at each value, and the value that minimizes the sum is adopted for every track on that CCD. Which tracks are flat depends on their fits, so the selection and σ_d are iterated to consistency, which takes one or two rounds.

2.4 Selection

The useful tracks are those which cross the full thickness of the sensor at nearly constant energy loss. We kept only the tracks for which the fit constrains t_c to within 0.15 and improves on a model with no diffusion by at least ten in M, for which the median charge per slice in the last third of the track is within 50% of that in the first third, since a rise in the deposited charge along the track is the Bragg peak of a particle that stopped inside the sensor and did not cross its full thickness, and for which the fit finds a depleted end, $t_c \leq 0.7$, since a particle that crossed the sensor is pixel-sharp where it left through the gates, whereas a feature that is wide from end to end is, on the slit-jaw imager, usually a spicule or other structure at the limb that the mask let through. We call these the flat tracks; 4584 tracks pass these cuts, 1049 of them on the SJI CCD, and Figure 10 shows the longest of them alongside a stopping track.

3 Results

3.1 The width against depth

Figure 4 shows the flat tracks stacked on their fitted centerlines and the diffusion width they imply at each depth, measured two ways. The first assumes no depth dependence at all: for each depth bin, the misfits of every slice in the bin are summed over a grid of trial widths and the width which minimizes the sum is taken, so the wedge shape emerges from the data rather than being imposed. The second applies the same procedure to the fractions the per-track fits of the slab model predict for the same slices, so that the fits are measured the way the slices are, although their cloud is not a Gaussian. The two agree to within 0.30 μm on all four CCDs at every depth but the bins just past t_c, where the measured cloud is up to 0.75 μm wider than the fits: it narrows less abruptly at the

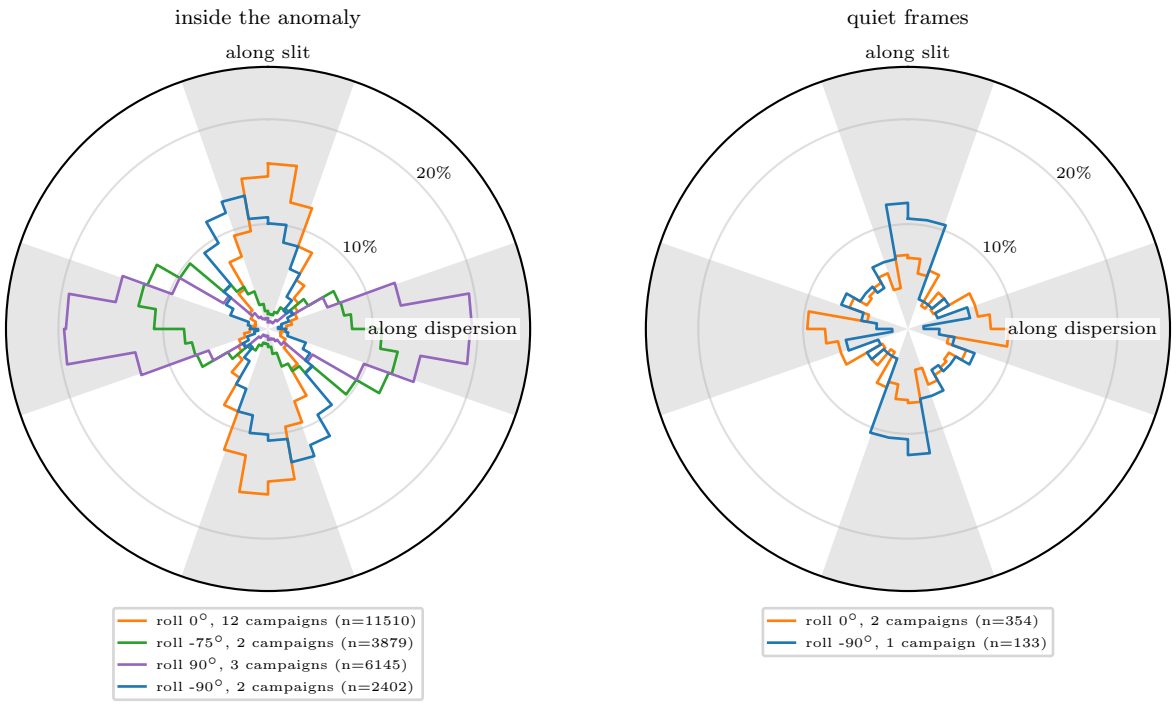


Fig 2 The azimuth in the detector frame of every elongated connected component above 5σ in the spectrograph frames of every campaign, with a charge-weighted width under 3 pixels and an aspect ratio over 4, before any of the finder's cuts, so that the whole circle is populated, grouped by the roll of the spacecraft. Tracks are undirected, so each is drawn at θ and $\theta + 180°$, and the grey wedges are the windows about the row and column axes that the finder accepts. Left: the frames taken inside SAA. Right: quiet frames. The trapped protons arrive strongly aligned, along the slit at rolls of $0°$ and $-90°$, along the dispersion at $90°$, and between the two at $-75°$, so the preferred direction is a property of the arrival direction as seen by the spacecraft rather than of the detector. The cosmic rays of the quiet frames at roll $0°$ are close to isotropic, while the quiet frames of the one campaign at $-90°$, taken at the edge of the anomaly, keep some of its alignment.

edge of the depletion region than either model has it. Beyond t_c the model-free profile does not go to zero but keeps a floor across the depletion region, under a micron just past t_c and a quarter micron near the gates, and the fits follow it: this floor is the σ_d term of Equation 2. The third panel shows the kernel itself on FUV2 at four depths, integrated over a pixel: the fits follow its core as it narrows from the back surface to the gates, while the stray charge discussed with Figure 8 lowers the measured peak by a few hundredths and raises the wings.

3.2 The shape of the charge cloud

Table 2 and Figure 5 compare the slab model with the Gaussian of Equation 3, each fit to every track and its σ_d pooled over each CCD in the same way; 4584 tracks are flat under the slab model and 4387 under the Gaussian. On the 4088 tracks flat under both, the slab model lowers the summed misfit by 12,018, 2.9 per track, and it does so on every CCD, fitting 51% of the tracks better by more than two and 20% worse. The gain is in the field-free layer, between the back surface and the critical depth, where the two clouds differ (Figure 5b); beyond t_c, where both are the same Gaussian, the slab model is a little worse, by about a tenth per track in each depth bin, since the

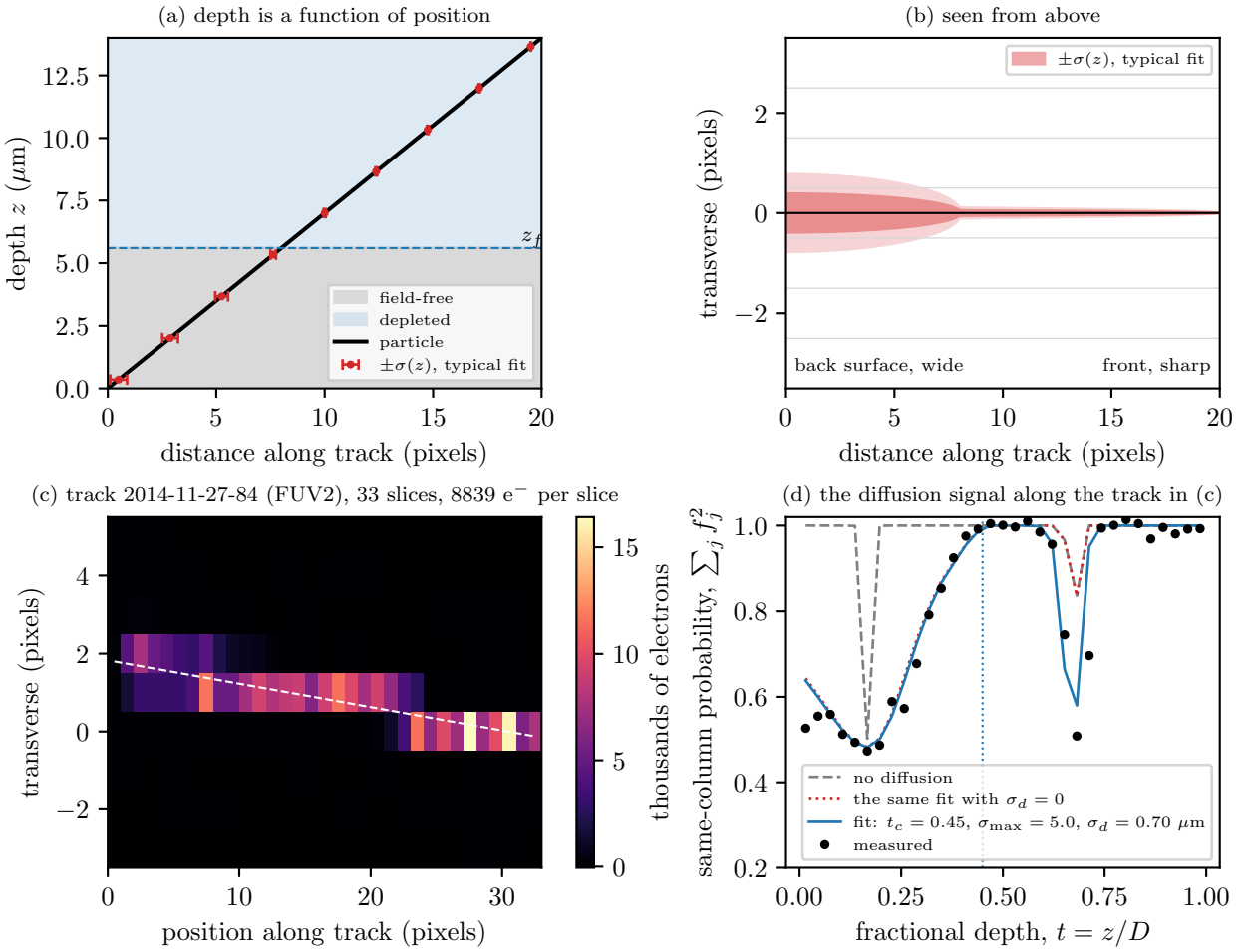


Fig 3 How a glancing track measures the diffusion width against depth. (a) A particle entering at the illuminated back surface ($z = 0$) and leaving at the gates ($z = D$) crosses the field-free layer first, where the charge it liberates diffuses until it reaches the depletion edge, so the lateral spread (red bars, to scale in pixels, at the median fit of the FUV2 tracks and its σ_d) is largest at the back surface and falls steeply to $z = z_f$; beyond it only the drift across the depletion region spreads the charge, by under a micron, narrowing to nothing at the gates. (b) Seen from above, the track is a wedge: about 0.4 pixels wide at one end and a few hundredths of a pixel beyond t_c. (c) A proton track on the FUV2 CCD, in the coordinates of its parent image, with the fitted centerline dashed. (d) The probability that two electrons deposited in the same slice of that track are collected in the same column, $\sum_j f_j^2$ corrected for read noise, for every slice, against the best fit, the same fit with $\sigma_d = 0$, and a model with no diffusion, all evaluated at the fitted centerline. The dips where the no-diffusion curve falls below one are where the centerline crosses a column boundary; the diffusion signal is the gap between the measured points and that curve. Beyond t_c the cloud is under a micron wide, a small fraction of a pixel, and changes $\sum_j f_j^2$ only where the centerline runs within a tenth of a pixel of a boundary, as it does near $t = 0.65$: there the measured dip follows the fit with σ_d and is deeper and wider than the fit without it.

centerline and the t_c it chooses serve the field-free layer. The gain grows with the charge per slice, which pins down the shape of the cloud, and at the same charge it is the same for the 8 and 15 s exposures and for the protons and the cosmic rays, so it is not the stray charge of Section 3.5, which grows with the exposure and is absent from the frames of the cosmic rays. It is the shape of the cloud that the tracks prefer rather than the way its width falls with depth: a Gaussian whose

variance falls as Equation 2 has it fits the same tracks worse than Equation 3, by 2,453. Tying $\sigma_{\max}$ to the thickness of the field-free layer, $t_c D$, as the slab model has it for a sensor 14 μm thick, costs 23,411, more than the shape gains, for reasons the next sections take up. The model-free widths summarize each depth bin by a single Gaussian and do not separate the models as sharply: under both, the residuals of Table 2 are a few tenths of a micron, but at the back surface the measured width falls below the Gaussian fits on every CCD, by up to 0.55 μm, and within 0.25 μm of the fits of the slab model, whose variance stops changing with depth where the back surface reflects the charge.

3.3 The spread inside the depletion region

Figure 6 shows how σ_d is determined. The misfit M of Equation 4 summed over the flat tracks on each CCD has a sharp minimum, at $\sigma_d = 0.80$, 0.70, 0.80, and 0.80 μm on FUV1, FUV2, NUV, and SJI, with the neighboring grid points a tenth of a micron away disfavored by 27 to 730 units of misfit. The improvement over $\sigma_d = 0$ is 1,613 to 3,402 units pooled but 1.8 to 2.6 per track, and the value each track prefers on its own is spread over the whole grid, from 0 to 1.5 μm, so the effect is modest for any one track and consistent across them, which is why σ_d is fit to each CCD as a whole. The resolution of the centerline matters here: a centerline located only to 0.1 pixel is uncertain by 1.3 μm, the size of the effect being sought, and with such a grid the minimum washes out entirely. σ_d trades against the field-free wedge: as it rises from zero to its adopted value, the mean t_c of the flat tracks falls by 0.02 to 0.05, most on FUV1, since without it the wedge stretches to absorb the floor, while the mean $\sigma_{\max}$ rises by 1.6% at most. Leaving σ_d out of the model would therefore bias the critical depth, which is why it is part of the model rather than a correction applied afterwards.

3.4 The critical depth and back-surface width

Figure 7 shows the fitted t_c and $\sigma_{\max}$ of every flat track, and Table 1 their means in each campaign. Half of the flat tracks have a critical depth between 0.35 and 0.45. Its mean over the core tracks is the same to within 0.02 for the two particle populations, the four spacecraft rolls, and the 8 and 15 s exposures, and every campaign with more than twenty core tracks lies between 0.37 and 0.40; the four CCDs differ from one another by up to 0.04, FUV1 lowest and FUV2 highest. The back-surface width differs more between chips, its median from 4.0 μm on FUV1 and 4.5 μm on NUV to 5.0 and 5.0 μm on FUV2 and SJI, so the pooled scatter is mostly this chip-to-chip difference, which is expected from the resistivity of the wafer and the applied bias and is one reason a sensor's kernel is worth measuring rather than assuming.

The ratio of the two measures the thickness of the sensor, since the cloud at a reflecting back surface is as wide as the field-free layer is thick, $\sigma_{\max} = z_f = t_c D$. The mean $\sigma_{\max}$ of the flat tracks divided by their mean t_c is 11.0, 12.8, 11.7, and 12.4 μm on FUV1, FUV2, NUV, and SJI, each to about 0.1 μm, against the 14 μm of the design.

3.5 The same-pixel probability

For the noise statistics of a UV image the quantity that matters is not the width but the probability that two electrons deposited at the same depth are collected in the same pixel. For a slice with pixel fractions f_j that is $p = \sum_j f_j^2$, corrected for the read-noise contribution $\sum_j \epsilon_j^2$, and it can be read directly off each slice with no model of the shape of the kernel. For a separable kernel

the probability that two electrons are collected in the same pixel would be p^2, but the cloud of the slab model is not separable, since an electron that takes long to reach the depletion region spreads along both axes at once. For the median fit of each CCD at the back surface the slab model puts the same-pixel probability 1.5 to 2.4% above p^2, and we take $\mathcal{P}$ to be p^2 raised by that factor. Figure 8 shows p against depth on each CCD, and Table 3 lists it for the slices within $D/10$ of the back surface. On the SJI CCD the back surface gives $\mathcal{P} = 0.348 \pm 0.004$, FUV2 the same to within the errors, while NUV and, further still, FUV1 spread their charge over fewer pixels, in the same order as their back-surface widths.

The figure gives p two ways in each depth bin, as the plain mean over the slices and as the median of how far each slice falls short of its own sharp-cloud value. The two agree at the back surface and part beyond t_c, where the mean settles near 0.88 on all four CCDs while the median stays within 0.02 of the 0.95 of a perfectly sharp cloud, with the per-track fits between them. The difference is charge that does not belong to the track. A frame taken inside the anomaly is crowded with hits, and one that touches a track is joined to it by the finder and cut out with it. Such charge lands only off the peak of a slice, so it can only lower p, and it pulls the mean down in proportion to its charge while the median is unmoved until it reaches half the slices. Two checks say this is what the mean is seeing: it falls twice as far below the sharp-cloud value in campaigns exposed for 15 s as in those exposed for 8 s, at the same charge per slice, and it does not fall at all for the few tracks of cosmic rays recorded outside the anomaly, where the frames are empty. At the back surface the cloud already spans several columns and the same stray charge moves p by under 0.02, so Table 3 keeps the mean, which is the quantity that enters the variance of an image. Beyond t_c the median is the measurement, and it leaves the depletion region a few hundredths short of sharp, in the direction and about the size of the σ_d the fits carry.

Table 2 The slab model of Equation 2 against the Gaussian model of Equation 3, on the tracks flat under both. ΔM is the misfit of the slab model minus that of the Gaussian, summed over the tracks, each model at its own σ_d, and the next two columns count the tracks the slab model fits better or worse by more than two. The residual is the model-free width of Figure 4b minus that found the same way in the fractions the fits predict, in the depth bin at the back surface and as the root mean square over all twenty bins, for the Gaussian model and then the slab model.

CCD	tracks	ΔM	better	worse	back residual (μm)	rms residual (μm)
FUV1	905	-846	358	249	$-0.55 \rightarrow -0.15$	$0.19 \rightarrow 0.13$
FUV2	1493	$-6{,}685$	894	251	$-0.30 \rightarrow +0.25$	$0.20 \rightarrow 0.21$
NUV	731	$-1{,}203$	307	165	$-0.15 \rightarrow +0.20$	$0.11 \rightarrow 0.22$
SJI	959	$-3{,}285$	515	157	$-0.45 \rightarrow +0.15$	$0.19 \rightarrow 0.15$

4 Discussion

At the back surface, where UV photons are absorbed, the tracks give the kernel directly: on SJI the field-free layer is 0.40 of the thickness, the cloud is 5.0 μm wide, and the same-pixel probability is 0.348 ± 0.004, the quantity that enters the variance of a UV image. Deeper in the sensor the tracks see a spread of under a micron, σ_d, that a purely field-free kernel would neglect. It is about what the drift-time diffusion across the depleted thickness plus the micron-scale ionization column of the track itself should give, and it is small compared with the back-surface spread, so it has little effect on UV imaging. It cannot be left out of the fit, though: without it the wedge stretches to absorb the floor and t_c comes out too large, which is why Equation 2 carries it from the start.

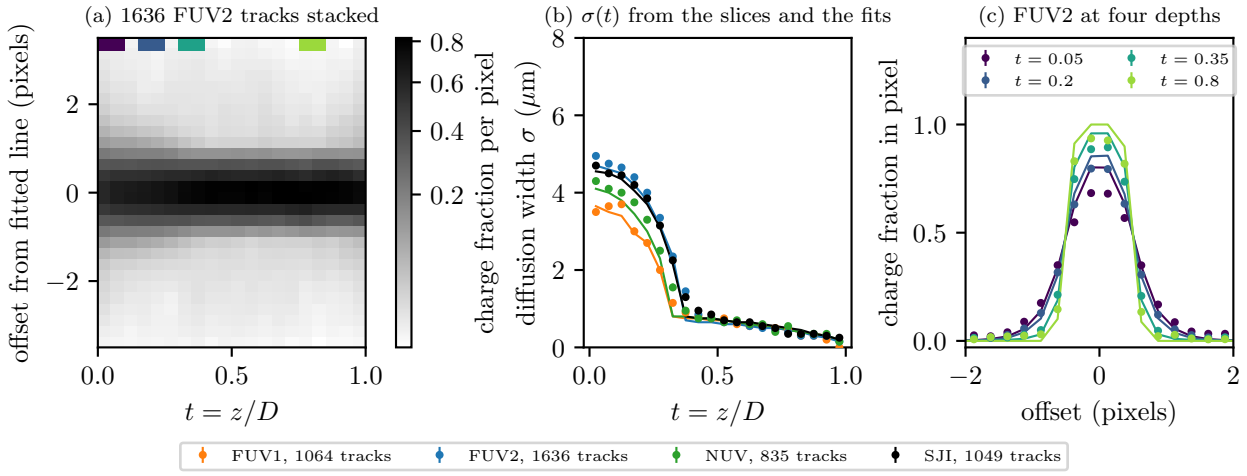


Fig 4 (a) Every slice of every flat track on the FUV2 CCD placed at its fractional depth and aligned on its fitted centerline, with each pixel's charge spread uniformly over its own width. The wedge is subtle: the wings at $t < 0.4$ carry a few percent of the charge. (b) Points: the diffusion width in each depth bin on each CCD, found by summing the misfit of Equation 4 over every slice in the bin on a grid of trial widths and minimizing, with no parametric model of the depth dependence; the error bars span the widths within two of the minimum M. Lines: the same procedure applied to the fractions the per-track fits of the slab model predict for the same slices, which follows the model-free profile at every depth, including the floor of a few tenths of a micron beyond t_c that the σ_d term supplies. The fits' cloud is not a Gaussian, so the width a Gaussian finds in it falls short of its standard deviation, Equation 2, most where the cloud has its narrow core, near the depletion region. (c) The kernel itself, integrated over a pixel: the mean fraction of the charge of the FUV2 slices in four depth bins, each a tenth of the thickness wide and marked along the top of (a), collected in a pixel whose center lies at the given offset from the fitted centerline, with its standard error. Lines: the same mean of the fractions the per-track fits predict for the same pixels. The measured peak falls a few hundredths below the fits at every depth, including the depletion region where the fits are pixel-sharp, and the measured wings stand above them: this is the stray charge that separates the mean from the median in Figure 8, which the robust misfit ignores and the mean includes.

Table 3 The charge diffusion measured from glancing particle tracks on the IRIS CCDs. The number of frames is the number of level-1 images searched, with the number taken inside SAA in parentheses. The flat tracks are those which constrain t_c to within 0.15 and show no Bragg rise along their length. σ_d is the spread inside the depletion region fit to each CCD as a whole, and t_c and $\sigma_{\max}$ are the medians of the per-track fits of the slab model at that σ_d, with the interquartile range in brackets, and $\bar{\sigma}_{\max}/\bar{t}_c$, the mean $\sigma_{\max}$ of the flat tracks divided by their mean t_c, is the thickness of the sensor they imply in μm, with its standard error. p is the probability that two electrons deposited within $D/10$ of the back surface are collected in the same column, and $\mathcal{P}$ is the corresponding probability that they are collected in the same pixel, p^2 raised by the factor the slab model gives for a cloud that is not separable.

CCD	frames	tracks	flat	σ_d (μm)	t_c	$\sigma_{\max}$ (μm)	$\bar{\sigma}_{\max}/\bar{t}_c$	p	$\mathcal{P}$
FUV1	8471 (2755)	2369	1064	0.80	0.35 [0.30, 0.40]	4.0 [3.0, 4.5]	11.0 ± 0.1	0.654 ± 0.004	0.435 ± 0.006
FUV2	8471 (2755)	3360	1636	0.70	0.40 [0.35, 0.45]	5.0 [5.0, 5.5]	12.8 ± 0.1	0.578 ± 0.003	0.343 ± 0.003
NUV	8474 (2755)	2036	835	0.80	0.35 [0.35, 0.40]	4.5 [4.0, 5.0]	11.7 ± 0.1	0.623 ± 0.004	0.396 ± 0.005
SJI	6847 (2750)	2279	1049	0.80	0.40 [0.35, 0.45]	5.0 [4.5, 5.5]	12.4 ± 0.1	0.583 ± 0.003	0.348 ± 0.004

The FUV1 CCD is narrower than the other three, at the back surface and in the fitted $\sigma_{\max}$ alike, and NUV sits between it and the FUV2 and SJI pair. We have no explanation beyond chip-to-chip variation of the field-free thickness, which depends on the resistivity of the wafer and the applied

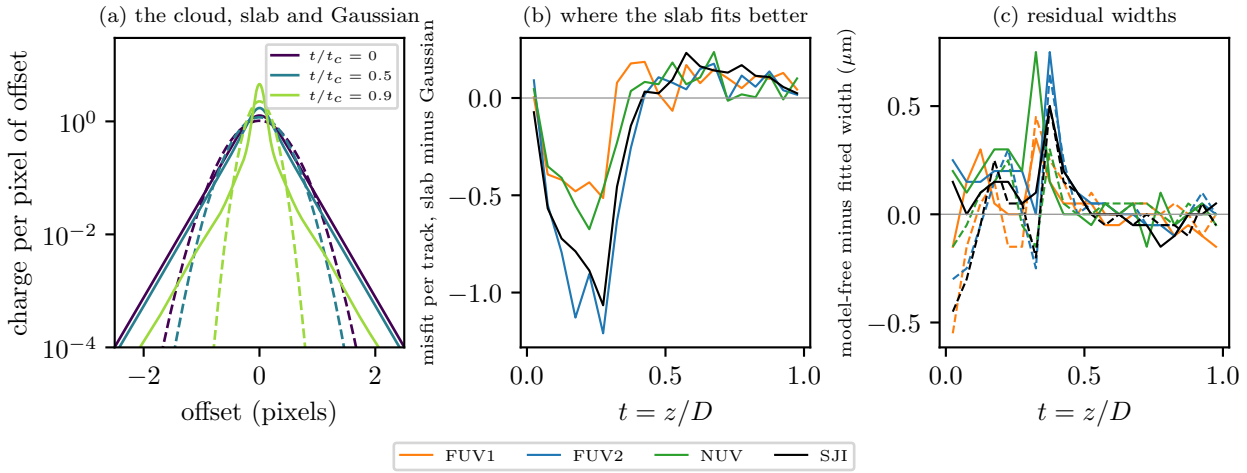


Fig 5 The slab model of Equation 2 against the Gaussian model of Equation 3. (a) The charge cloud along one axis at three depths in the field-free layer, for the median fit of the FUV2 tracks and its σ_d (solid), beside a Gaussian of the same variance (dashed). At the back surface the slab's cloud falls off exponentially rather than as a Gaussian, and near the depletion region it has a narrow core and broad wings. (b) The misfit of the slab model minus that of the Gaussian, per track, summed over the slices of each depth bin of the tracks flat under both models, each model at its own σ_d; below zero the slab model fits better. (c) The model-free width of Figure 4b minus that found the same way in the fractions the fits predict, for the slab model (solid) and the Gaussian (dashed).

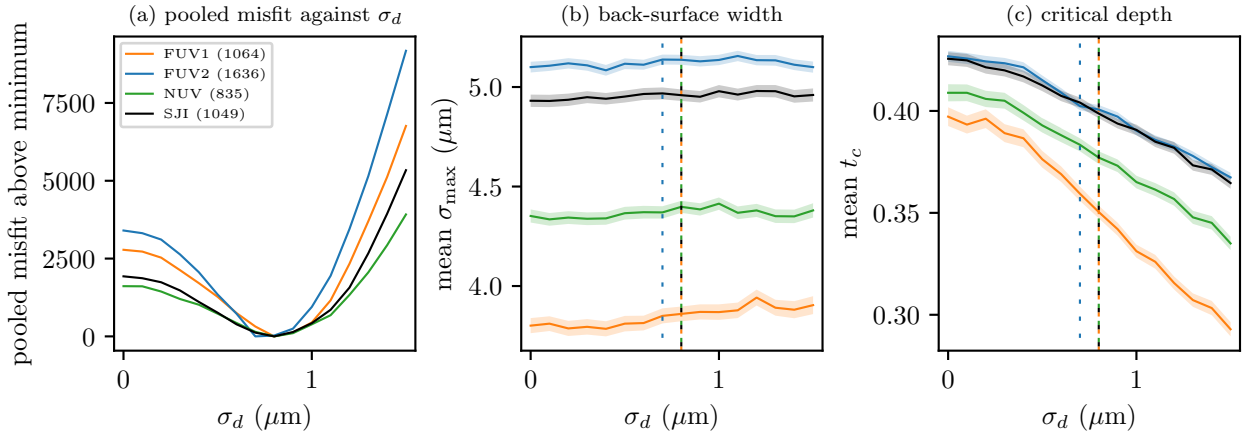


Fig 6 The spread inside the depletion region, σ_d in Equation 2, which is shared by every track on a CCD. (a) The misfit of Equation 4 summed over the flat tracks on each CCD at each σ_d, with each track refit in t_c, $\sigma_{\max}$, orientation, and centerline, shown relative to its minimum, which lies at $\sigma_d = 0.80$ μm on FUV1, 0.70 μm on FUV2, 0.80 μm on NUV, and 0.80 μm on SJI. (b) and (c) The mean $\sigma_{\max}$ and t_c of the same tracks at each σ_d, with the standard error of the mean shaded and the minimum of (a) on each CCD dashed: the spread inside the depletion region trades against the field-free wedge, so t_c falls steadily as σ_d rises, fastest on FUV1, while $\sigma_{\max}$ rises by 1.6% at most.

bias; the same measurement on the Atmospheric Imaging Assembly (AIA) CCDs, which are of the same design, would show whether such variation is common.

The slab model ties the width of the cloud at the back surface to the thickness of the field-free layer, so the tracks measure the thickness of the sensor, and they find it thinner than designed: 11.0 to 12.8 μm against 14, with FUV1 thinnest. Fixing it at 14 μm costs the fits more than the shape of

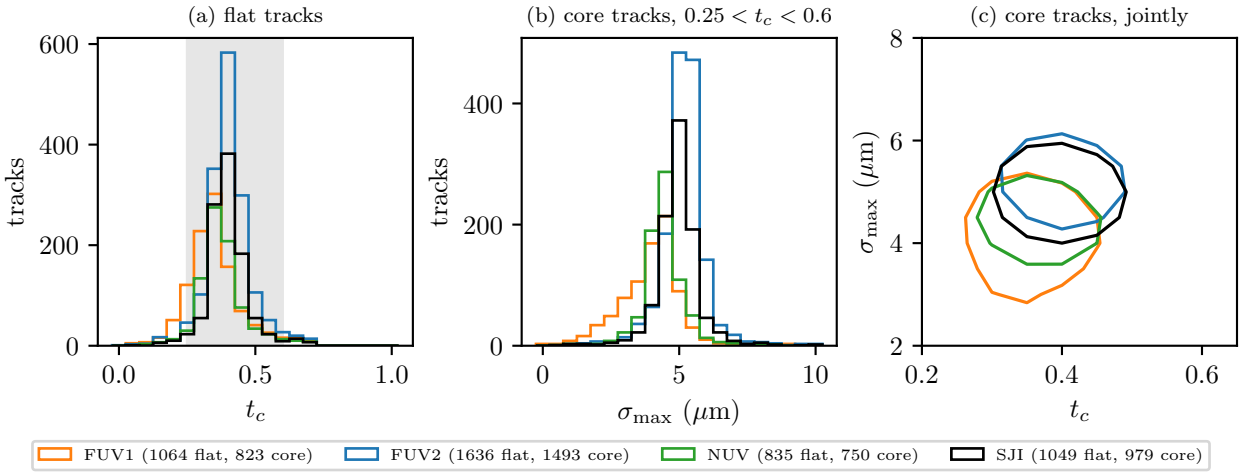


Fig 7 The fitted parameters. (a) The critical depth of every flat track on each CCD. The tails below 0.25 and above 0.6 are tracks Equation 2 describes poorly, sharp or diffuse along most of their length, and they are excluded from the shaded core used elsewhere. (b) The back-surface width of the core tracks. (c) The joint distribution of the core tracks on each CCD, as the contour enclosing half of them, on the grid the fits are searched over (0.05 in t_c and 0.5 μm in $\sigma_{\max}$) smoothed by one cell. The contours run wider in $\sigma_{\max}$ than in t_c, since a single track constrains its back-surface width less well than the depth at which it sharpens.

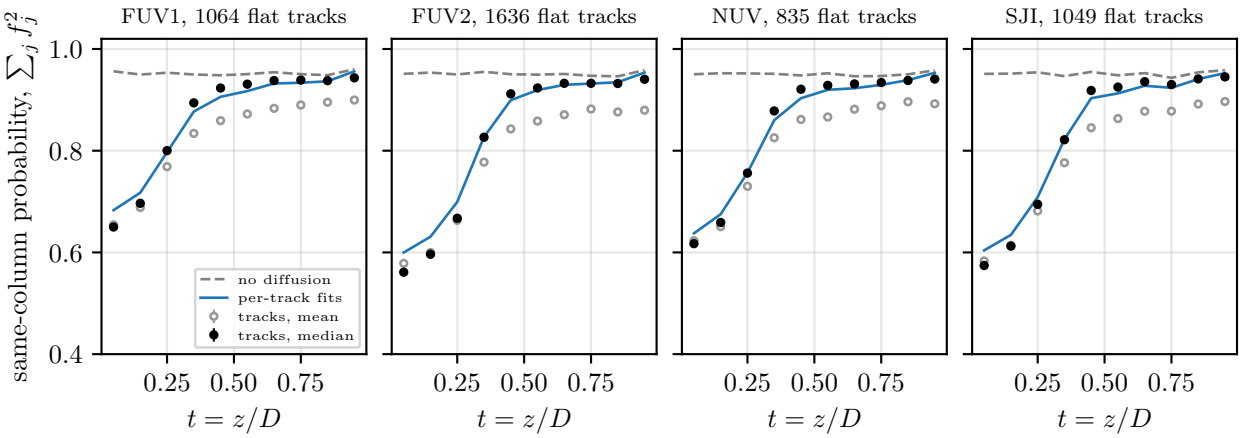


Fig 8 The probability that two electrons deposited in the same slice are collected in the same column, over the flat tracks on each CCD in bins of fractional depth. The dashed line is Equation 2 with no charge diffusion, averaged over the slices of each bin, which is below one only where a centerline runs near a pixel boundary, and the blue line is the same at each track's own fit. The black points are the dashed line minus the median, over the slices of the bin, of how far each slice falls short of its own no-diffusion value, with the standard error of the median; the open points are the plain mean of the slices. Charge from other hits in the frame that touched a track and was cut out with it can only lower the sum, so the mean sits below the median wherever such charge is common, beyond t_c on all four CCDs, while at the back surface the two agree. The back surface is lowest on the FUV2 and SJI CCDs and highest on FUV1.

the cloud gains them, so the tracks do not allow both the thickness of the design and a back surface that reflects all the charge reaching it. Either the sensors were thinned further than designed, or the back surface recombines part of the charge that reaches it,[1] which removes the electrons that would have spread the farthest and narrows the cloud most where it is widest. The two could be

told apart by a sensor of known thickness, or by the shape of the cloud near the back surface, which recombination changes and a thinner sensor does not.

Figure 9 turns the kernel into the quantity an instrument designer needs, the width of the charge cloud of a photon, by averaging it over the depth at which a photon of each wavelength is absorbed in silicon. Across the whole ultraviolet the absorption length is under 10 nm, so every photon sees the back-surface width of Table 3, a charge cloud 4 to 5 μm wide depending on the CCD. Only in the visible and the soft X-rays, where photons reach the depletion region, does the cloud narrow, and near the band gap the sensor is nearly transparent. The spread between the four CCDs is the spread a designer should expect between sensors of one design.

Three systematics deserve comment. First, the tracks are left by protons that deposit thousands of electrons per row, whereas a UV photon liberates one to three. In thick, fully depleted sensors, slow protons leave tracks visibly widened by the mutual repulsion of that charge,[12] and if the same occurred in the field-free layer here the tracks would overestimate the single-photon kernel. The tracks themselves can test this, by comparing the kernels of faint and bright tracks, and that comparison is the first thing to add. Second, the depth assignment assumes every flat track crosses the full thickness of the sensor; a track that entered or left through the edge of the field-free layer, or that was clipped by the Bragg cut at one end, would have its depths compressed. The agreement between the model-free and parametric profiles, and between campaigns, suggests such tracks are rare, but a validation on synthetic tracks injected into real frames is the way to bound it. Third, the stray charge of Section 3, from hits that touch a track and are cut out with it, is kept out of the same-pixel profile by the median but not out of the widths of Figure 4 nor the σ_d scan, both of which sum the misfit of every slice. It is one-sided and it grows with the exposure of the frame, so the floor of a few tenths of a micron beyond t_c and the σ_d of Equation 2 are upper bounds on the true spread, and the campaigns exposed for 15 s, a tenth of the flat tracks, carry twice their share of it.

5 Conclusion

Glancing particle tracks measure the depth-dependent charge-diffusion kernel of a back-illuminated CCD in orbit, with no laboratory access. On the IRIS CCDs they give the thickness of the field-free layer, the width of the charge cloud at the back surface, which differs by a fifth between four sensors of one design, and a spread of under a micron acquired inside the depletion region that a field-free kernel alone would miss. They also give the shape of the cloud, which is that of charge diffusing between a reflecting back surface and the depletion region rather than a Gaussian, and whose scale implies sensors thinner than designed.

Appendix A: The tracks as the finder cut them out

The cutouts distributed with this article are seven pixels wide and follow the fitted centerline, so they show the tracks after the finder has done its work. Figure 10 shows the longest of the flat tracks and one track that the Bragg cut rejects. Sub-pixel structure is invisible at this scale, which is why the wedge signature has to be extracted by fitting.

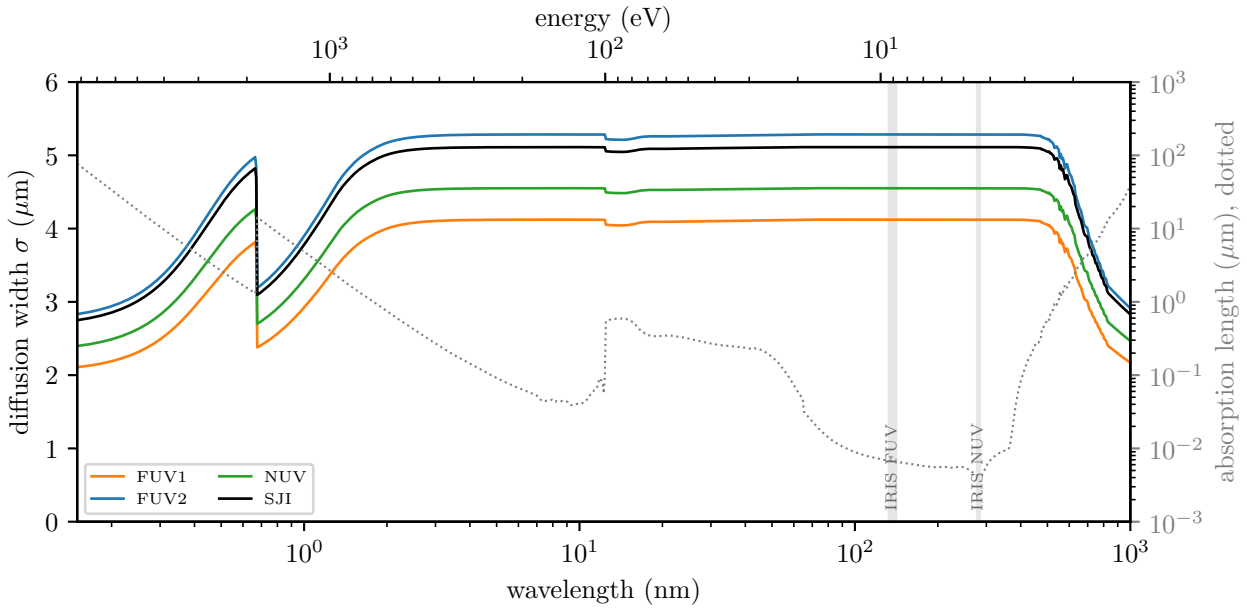


Fig 9 The width of the charge cloud a photon leaves in each CCD, against the wavelength of the photon. The width is the standard deviation of the cloud along one axis. A photon is absorbed at a random depth, exponentially distributed with the absorption length of silicon (gray dotted, from the tabulated optical constants[13,14]), and the charge it liberates spreads with the variance of Equation 2 at that depth, averaged over the per-track fits of the slab model on each CCD. The curve is the square root of that variance averaged over depth, weighted by the fraction of photons absorbed at each depth; photons that pass through the 14 μm of silicon are not counted. The fits rather than the model-free widths of Figure 4b give it, since the width a single Gaussian finds in each depth bin falls short of the standard deviation of a cloud with exponential tails. Between 30 and 350 nm every photon is absorbed within 10 nm of the back surface, where the variance of the cloud stops changing with depth, and the cloud has the root mean square $\sigma_{\max}$ of the fits. In the visible and in the soft X-rays the photons penetrate to the depletion region and the cloud narrows. Near the band gap silicon is nearly transparent, and the few photons absorbed are spread through its thickness.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper. The analysis code and the text of this article were drafted with the assistance of Claude, a large language model, under the direction and review of the authors.

Code, Data, and Materials Availability

This article is executable:[15] every figure, table, and number quoted in the text is computed from the tracks when the article is built. The code, the track cutouts, and the fits are available at `https://github.com/roytsmart/ccd-diffusion-paper`, and the level-1 IRIS images they were extracted from are available from the IRIS data archive.

Acknowledgments

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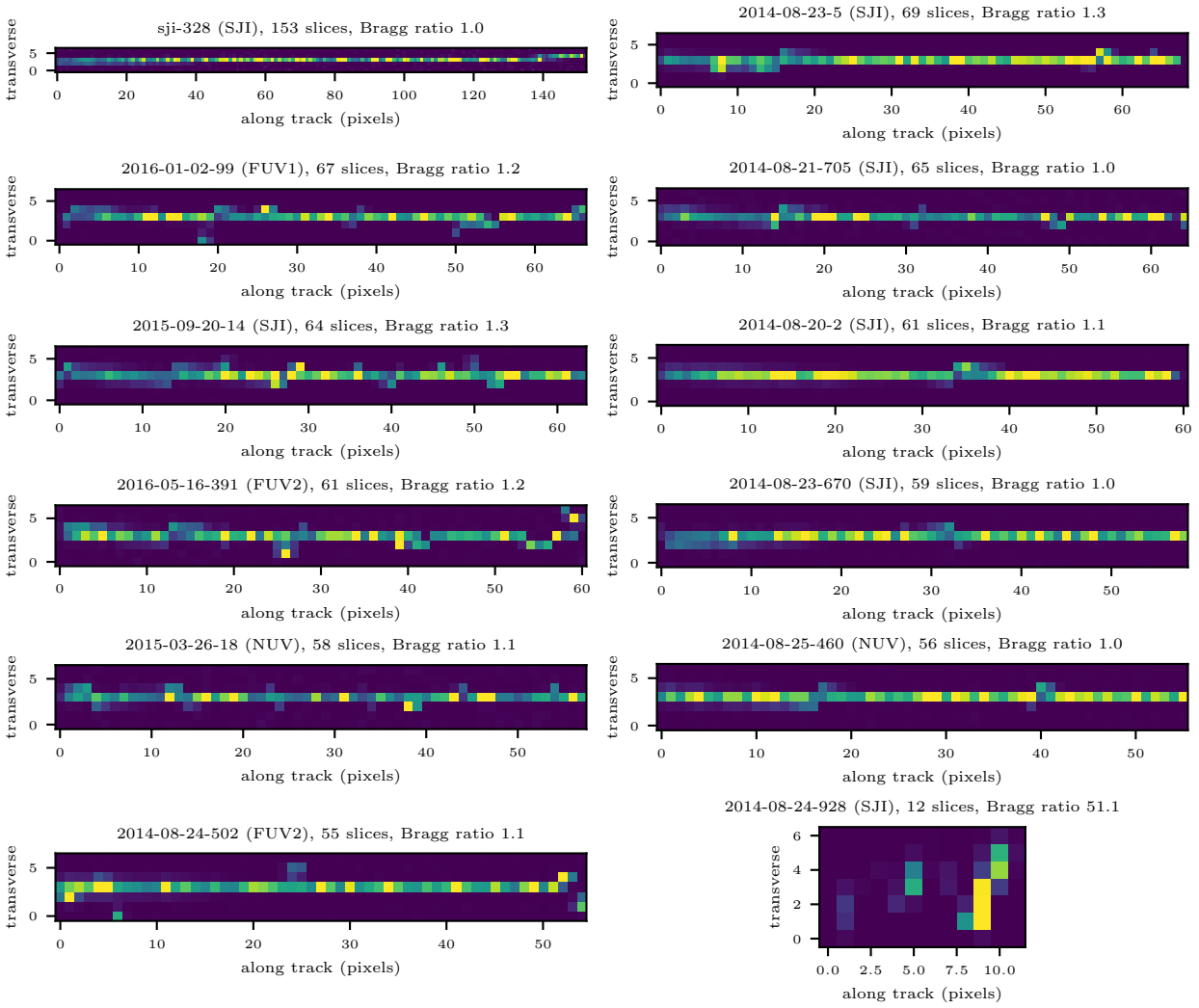


Fig 10 The eleven longest flat tracks and, in the last panel, the track with the strongest Bragg rise, as the finder cut them out: seven pixels wide, re-centered on the integer part of the fitted centerline in every slice, and oriented with the back surface at the left. Each panel's color scale is clipped at its 98th percentile so that a single bright pixel does not hide the track. The stopping track's charge per slice climbs by a factor of twenty toward one end, and the selection of Section 2 rejects it.

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